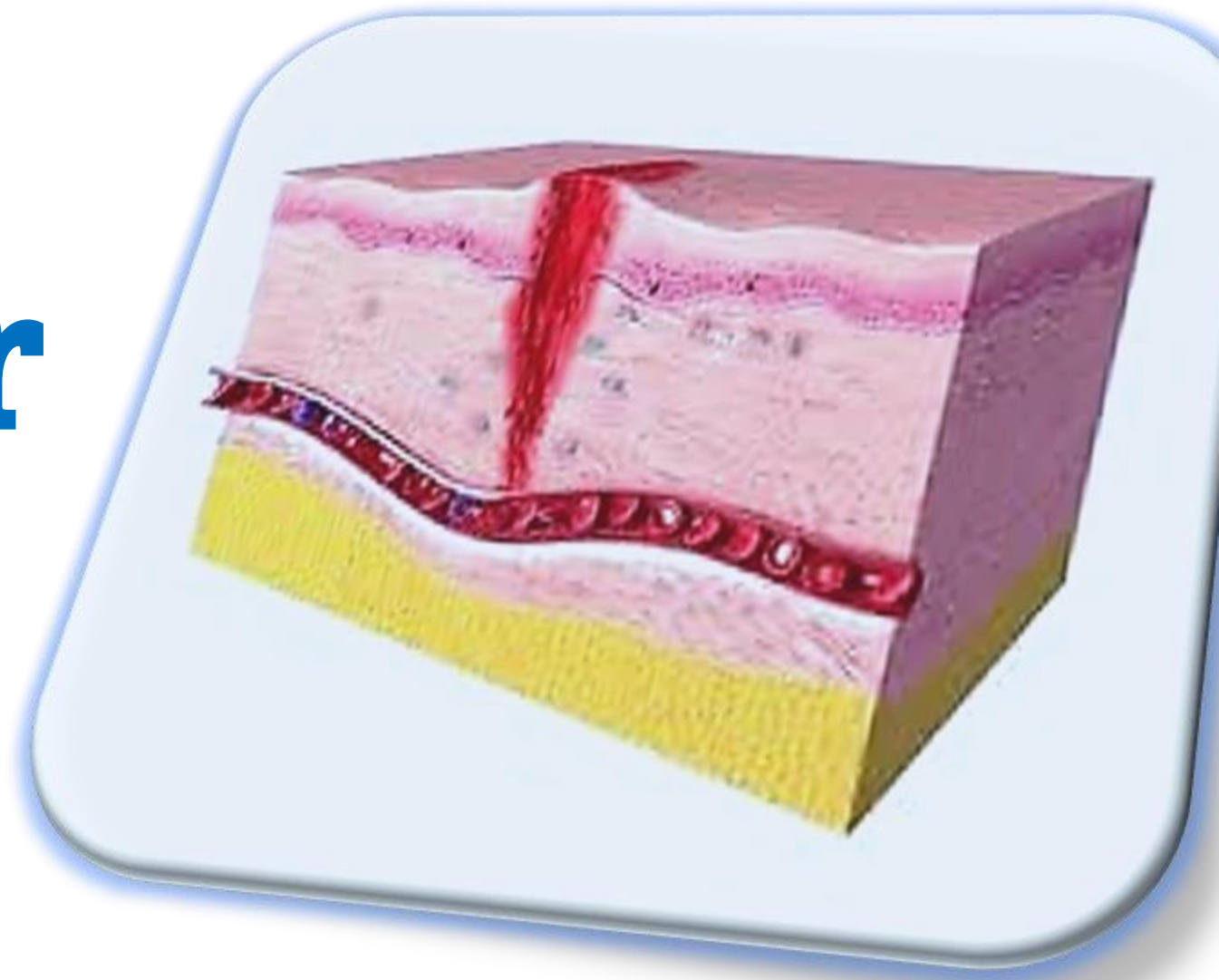


Healing and Repair

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Lecturer of pathology



Objectives

- Definition of repair or healing
- Differences between regeneration and scar formation
- Identification of types of tissue
- Mechanisms of tissue repair

REPAIR

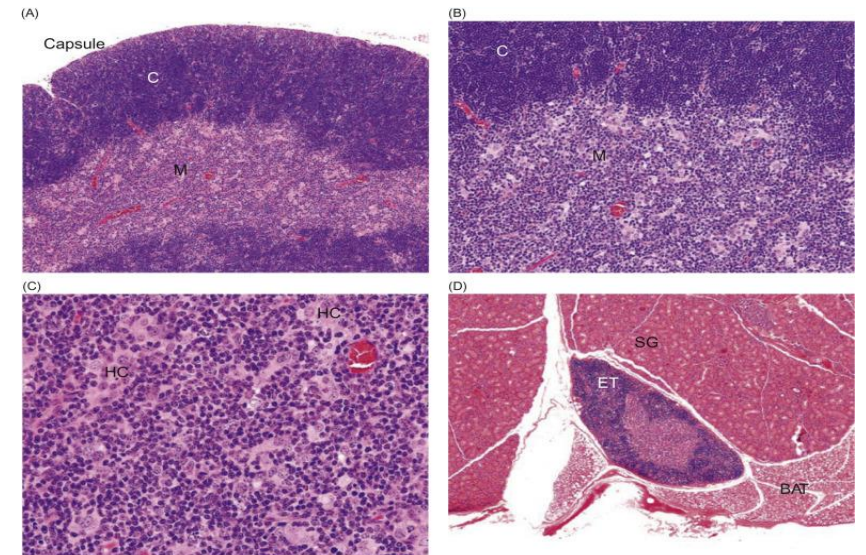
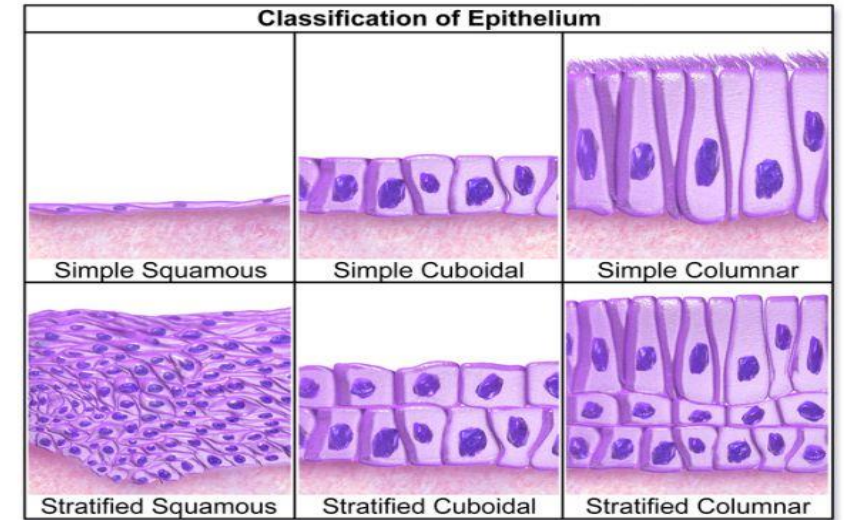
- Is restoration of tissue architecture and function
- **Regeneration:**
Replacement of damaged cells by new ones of the same kind (cells with capacity to proliferate)
- **Scarring:**
Production of a fibrous scar (cells without capacity to proliferate or tissue is severely damaged)

REGENERATION

- Cells are divided according to the power of regeneration into:

1- Labile cells:

- Proliferate continuously throughout life to replace aging cells
- Usually rapid regeneration
- They include →



REGENERATION

2- Stable cells:

- Not normally dividing at a significant rate, only on need.
- Speed of regeneration variable
- Small damage → regeneration.
- Large damage → repair with fibrosis
- Includes →

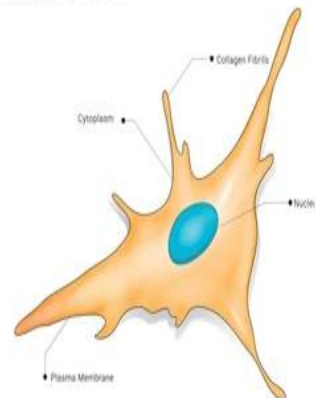


CHONDROCYTES



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Fibroblast Cell Structure



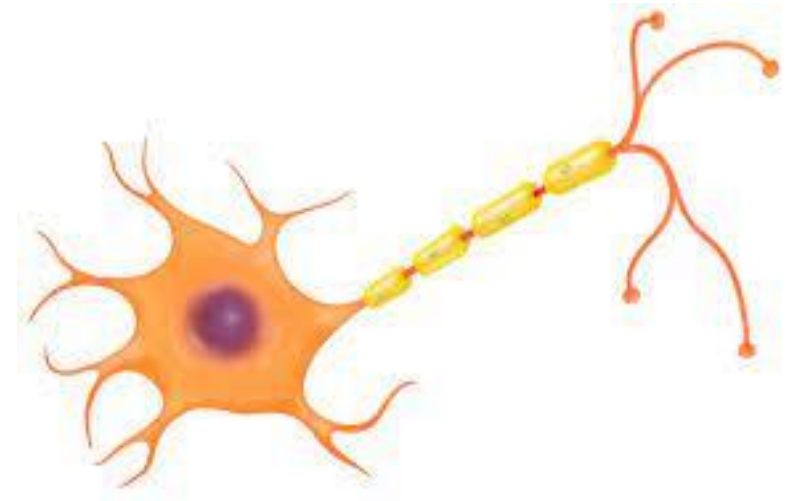
Fibroblast Cell

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REGENERATION

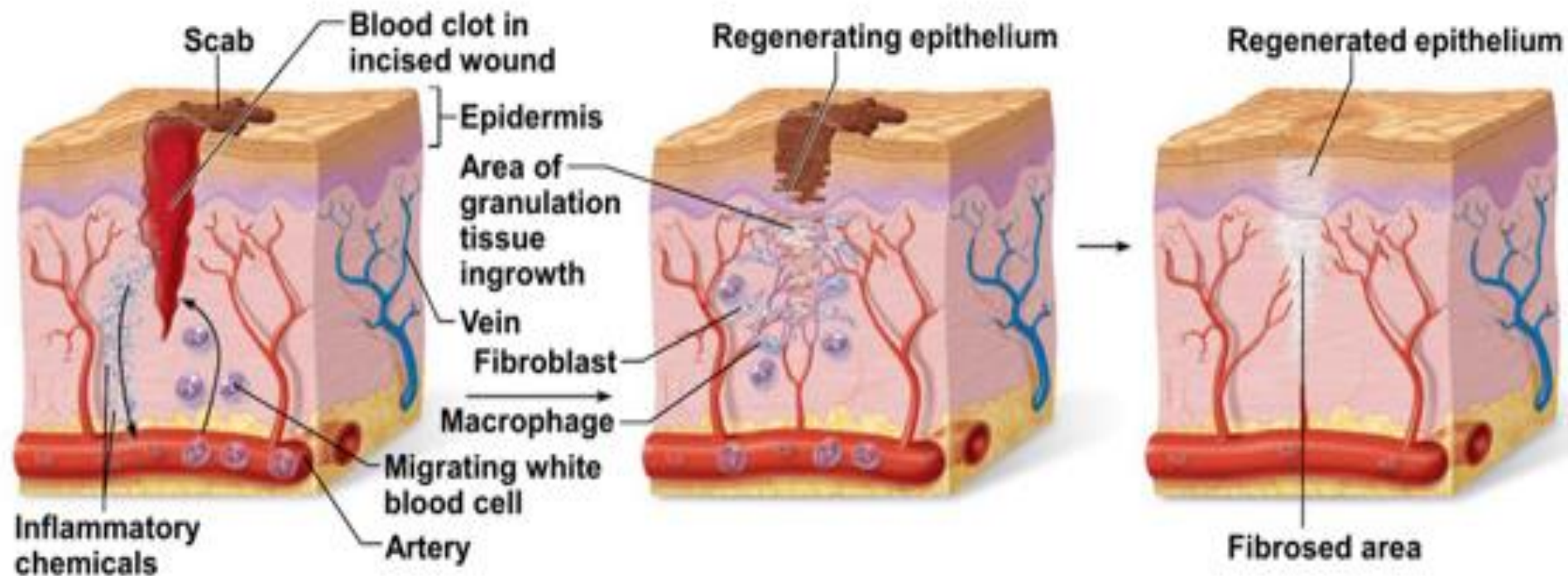
3- Permanent cells:

- Unable to divide
- Unable to regenerate
- Include →



Regeneration of epidermis and Mucous membrane

Epithelial cells at the margins proliferate and creeps from margins to the center until they fill the damaged covering then the basement membrane is formed.



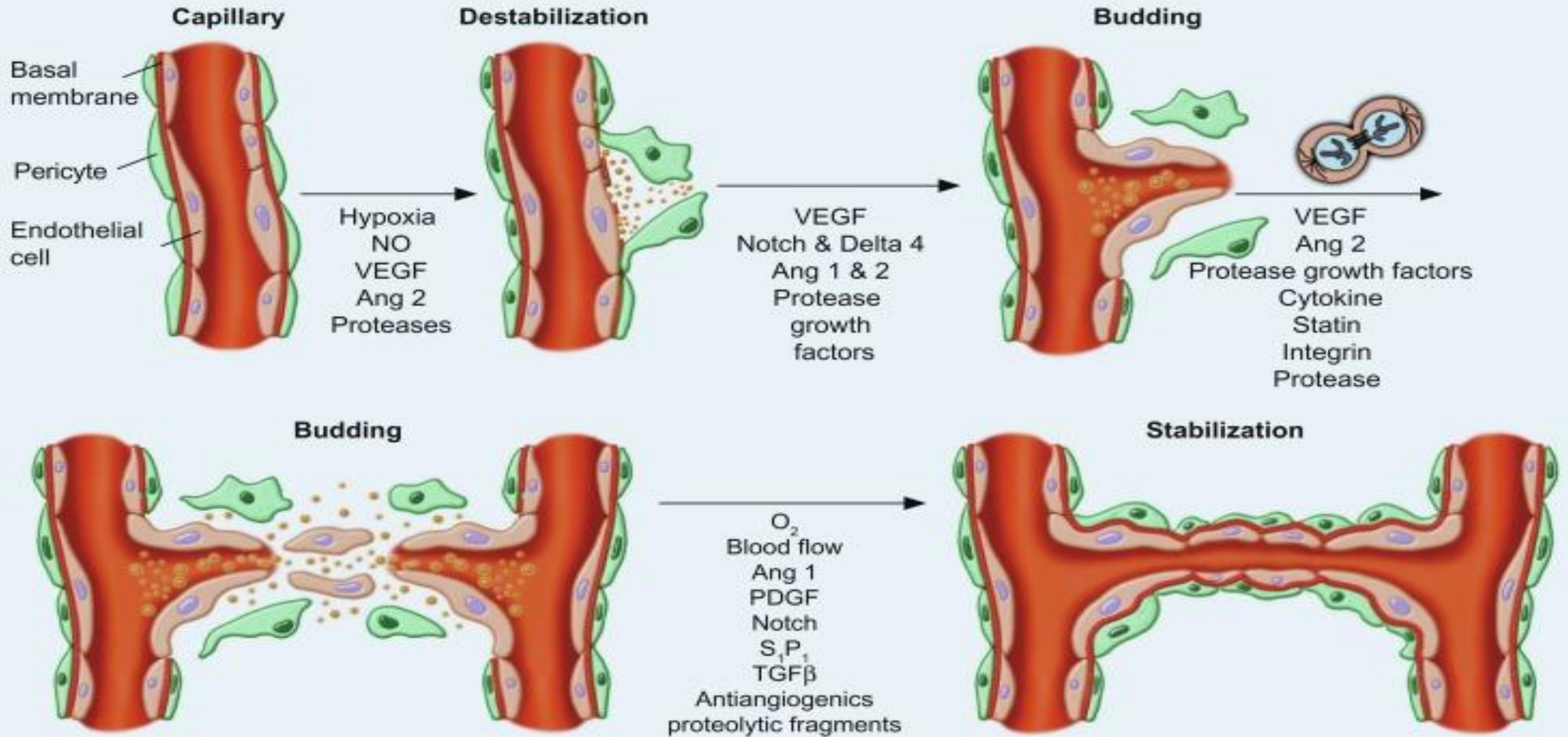
FIBROUS REPAIR

The development of a fibrous scar.

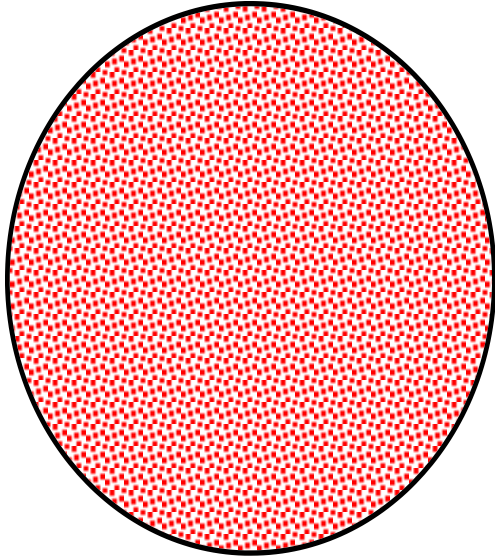
Fibrous repair

- Angiogenesis: (Vascular granulation tissue)
- Fibrogenesis: (Fibroblast migration, proliferation and collagen secretion)
- Maturation and remodelling

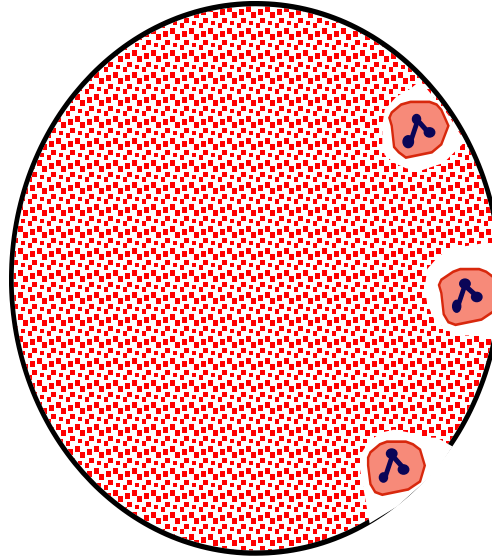
Angiogenesis



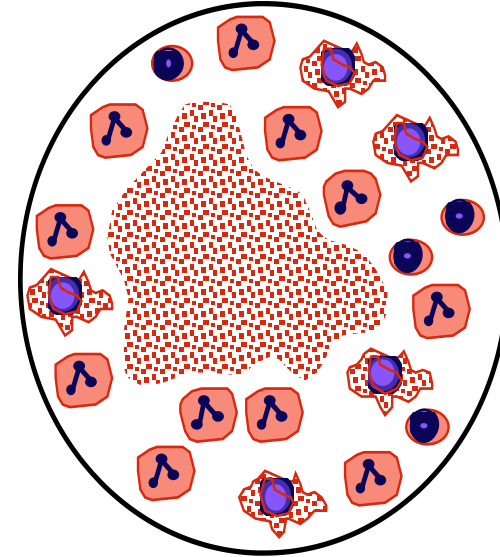
Steps of fibrous tissue repair



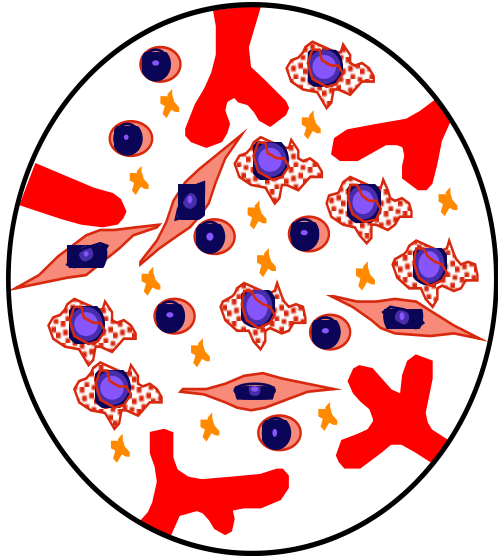
1) Exudate clots.



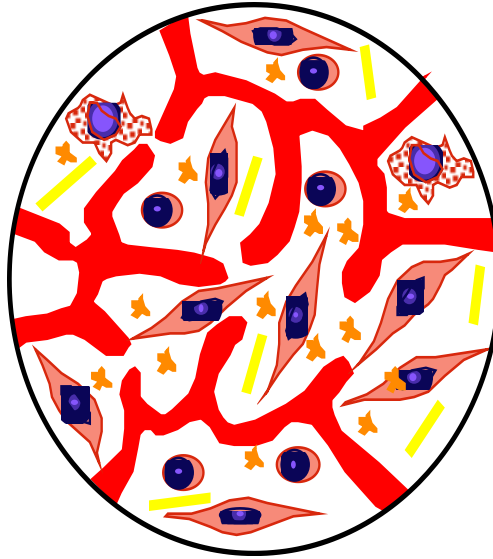
2) Neutrophils infiltrate and digest clot



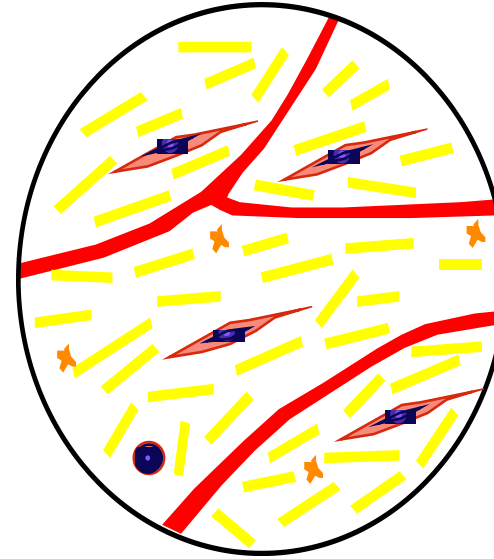
3) Macrophages and lymphocytes are recruited



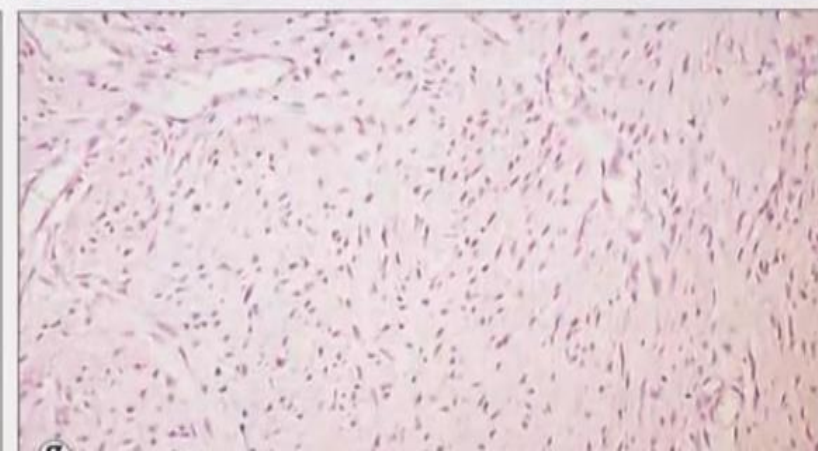
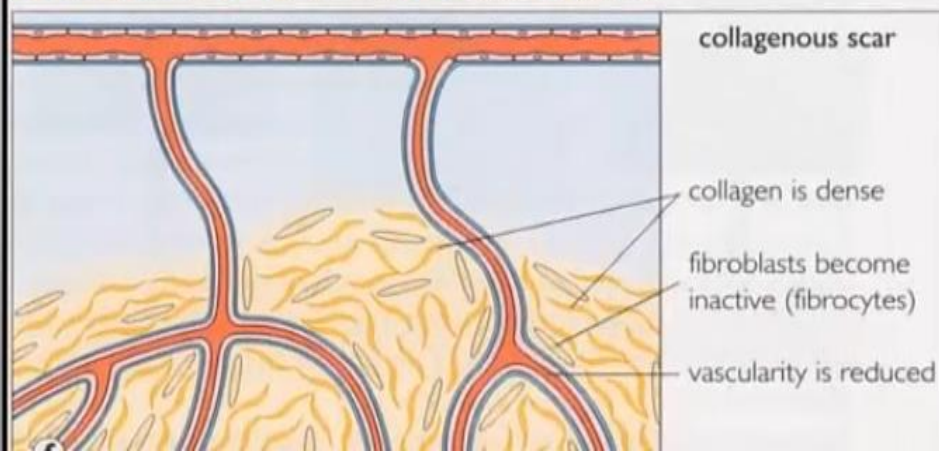
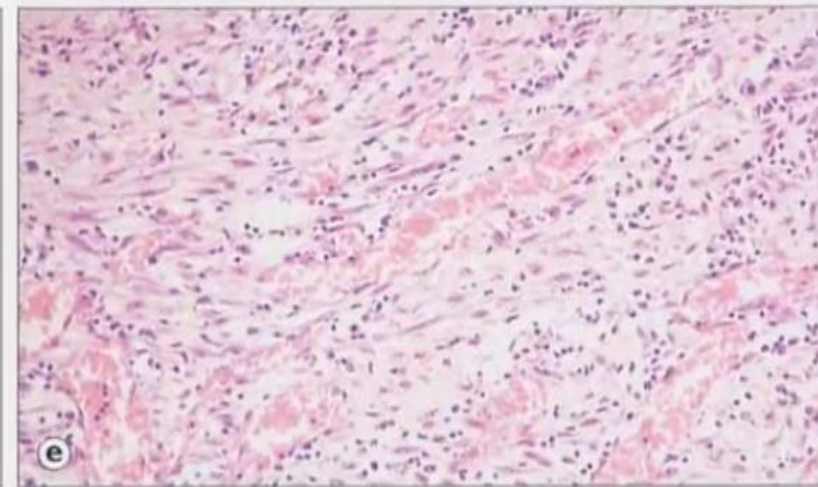
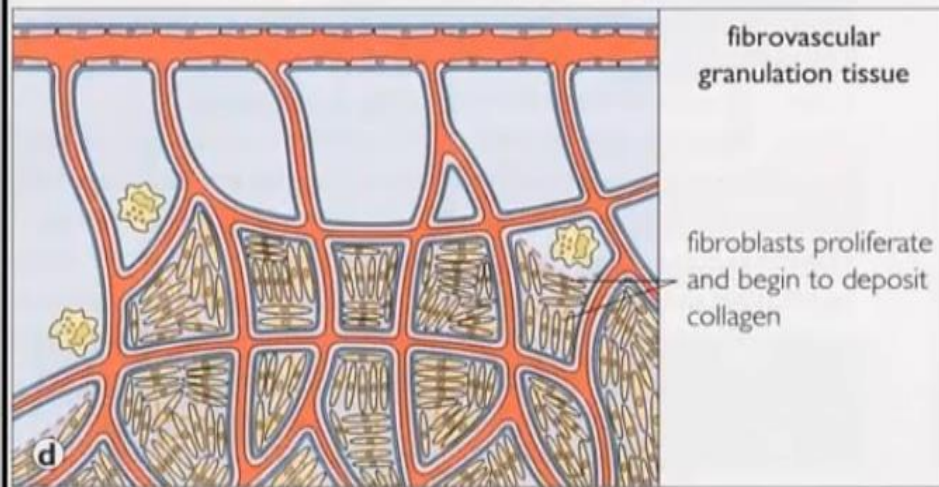
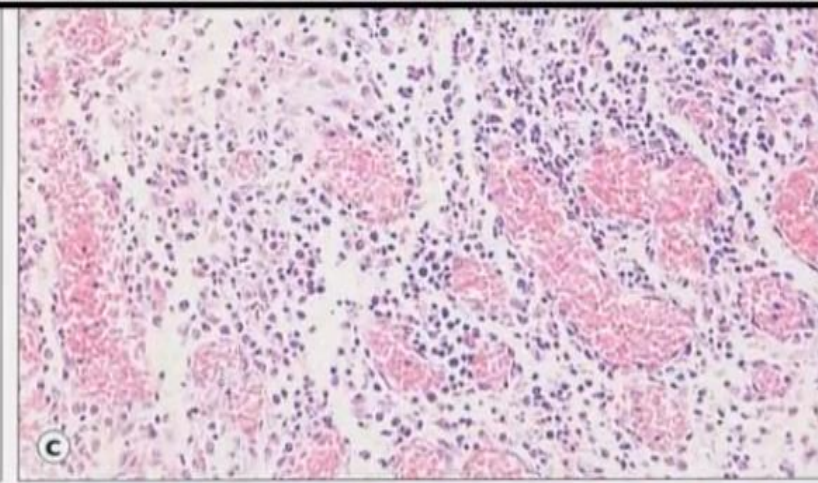
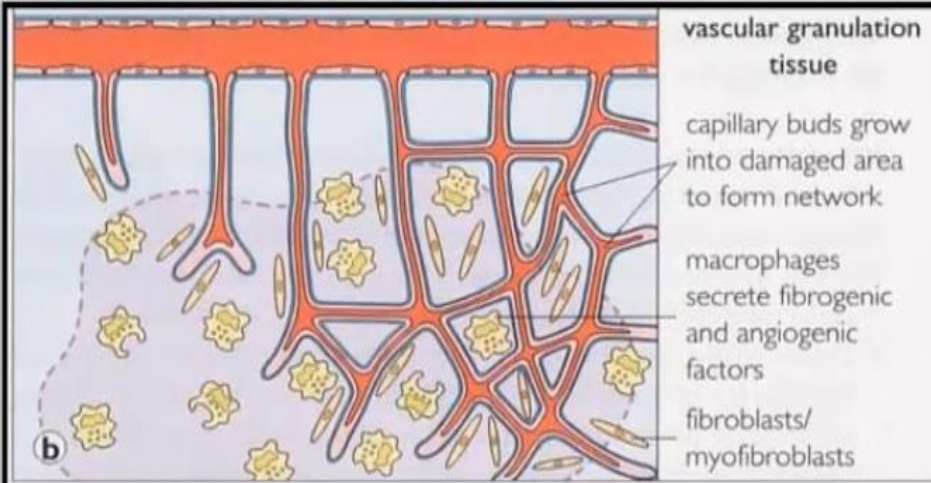
4) Vessels sprout,
myofibroblasts make
glycoproteins



5) Vascular network;
collagen synthesised;
macrophages reduced



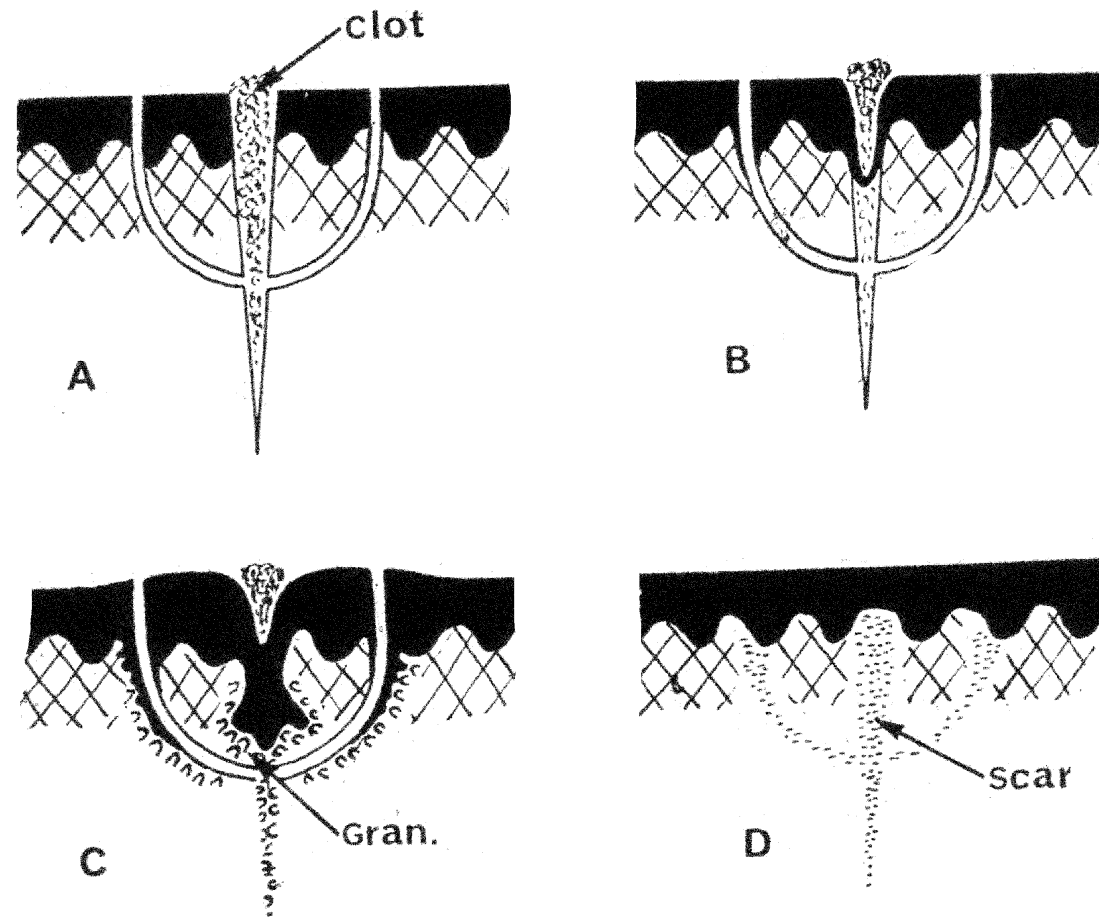
6) Maturity. Cells
much reduced; collagen
matures, contracts,
remodels



Skin Wounds

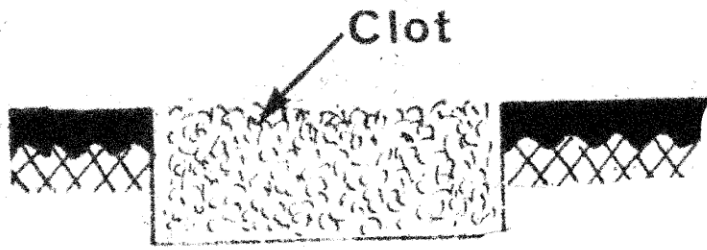
SKIN WOUNDS

Healing by "Primary Intention"

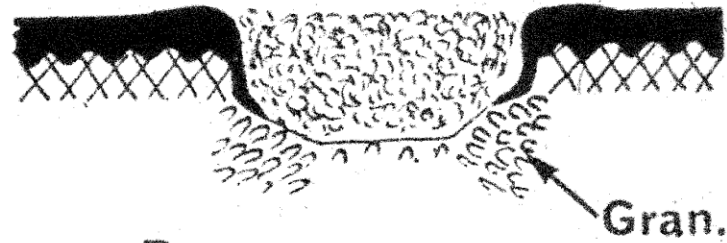


SKIN WOUNDS

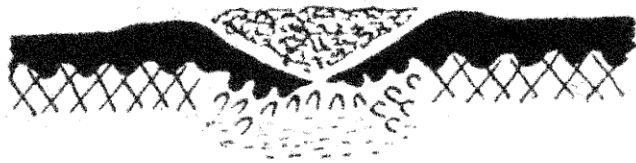
Healing by "Secondary Intention"



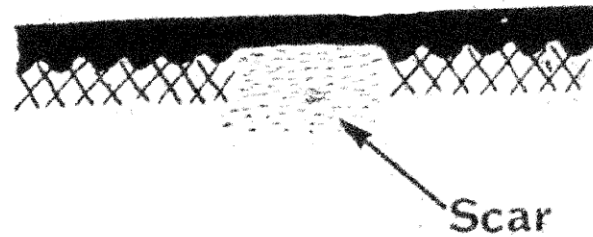
A



B



C



D

Differences between primary and secondary healing

Feature	Primary healing	Secondary healing
Cleanness	Clean	Unclean
Infection	Generally uninfected	May be infected
Margins	Surgically clean	Irregular
Healing	Scanty granulation tissue	Granulation tissue fill the gap
Healing period	Short	long
Healing direction	Direct healing	From the bottom to the edge
Outcome	Neat linear scar	Contracted irregular wound

FACTORS INFLUENCING WOUND HEALING

- **Local factors**

- Type, size, location of wound
- Apposition, lack of movement
- Infection: Suppuration, Gangrene, Tetanus
 - (Secondary hæmorrhage)
- Blood supply: Arterial, Venous
- Foreign material: dirt, glass, sutures, necrotic tissue
- Radiation damage

FACTORS INFLUENCING WOUND HEALING

- **General Factors:**

- Age
- General state of health
 - chronic diseases e.g. diabetes, rheumatoid arthritis etc.
- Drugs (n.b. steroids) and hormones
- General cardiovascular status
- General dietary deficiencies e.g. protein
- Specific dietary deficiencies
 - Vitamin C
 - sulphur-containing amino acids

Complications of Repair

- Insufficient fibrosis:
 - Ulceration, sinus and fistula
- Excessive fibrosis:
 - Cosmetic scarring; keloid
- Excessive contraction:
 - Limitation of joint movement (Contractures);
obstruction of tubes & channels (Strictures)

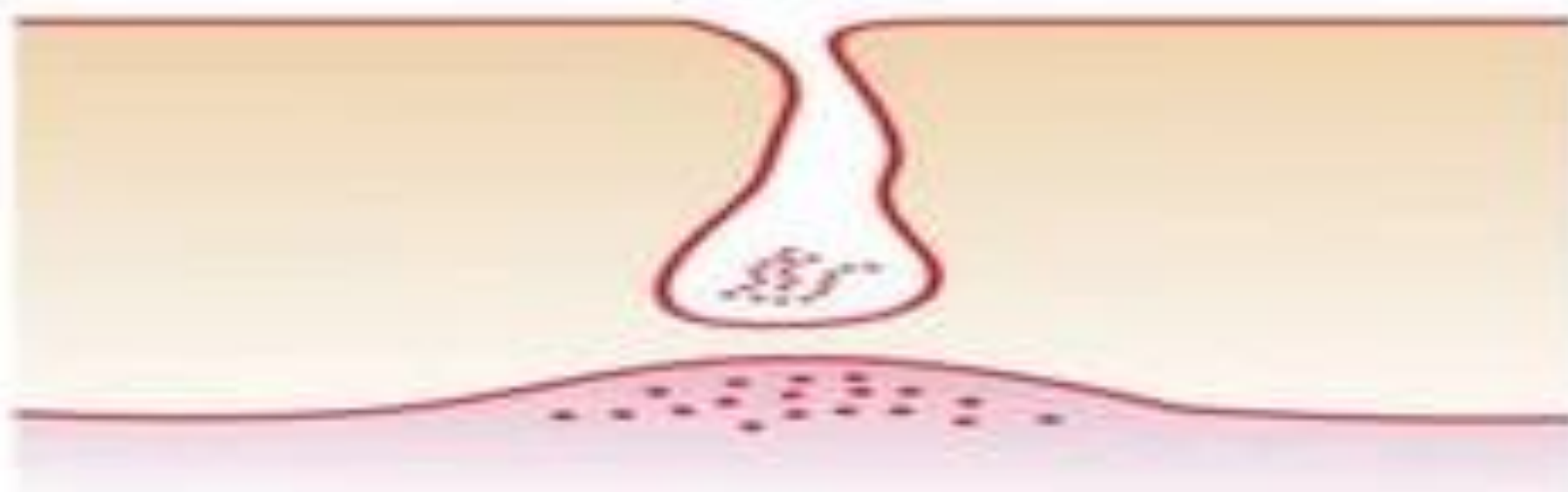
Chronic ulcer

Imperfect healing caused by persistent infection → destroys granulation tissue or excessive deposition of collagen at edges.



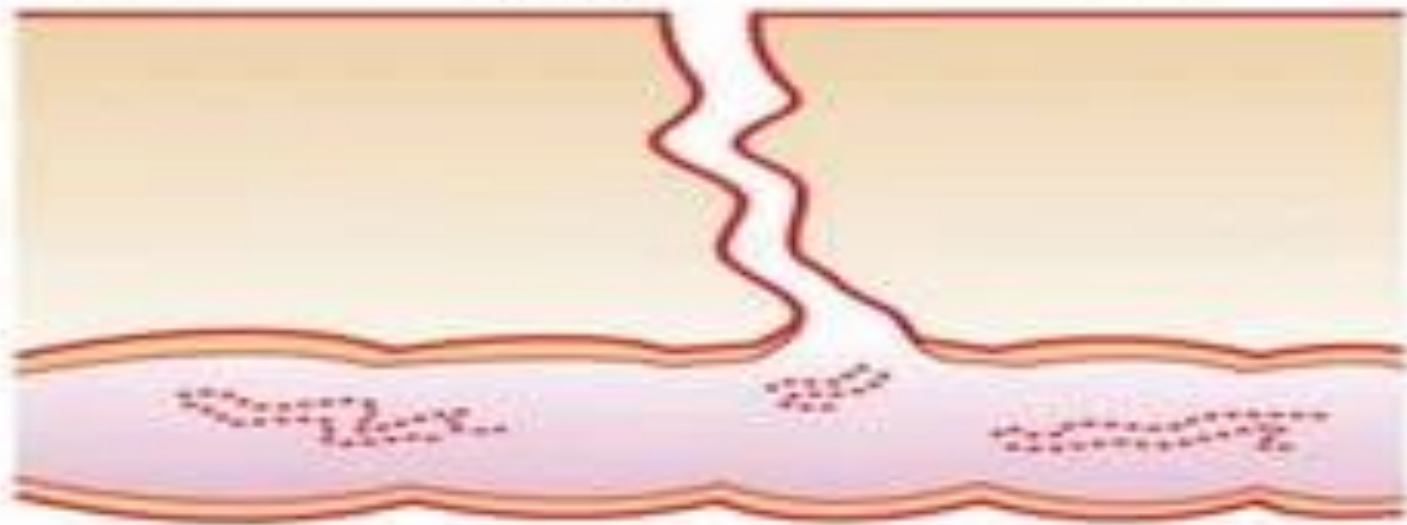
Sinus

- Def: Blind ending tract lined by granulation tissue leading from an epithelial surface down into the tissues.



Fistula

- Def: A fistula is an abnormal communication between two epithelium lined surfaces. This communication or tract may be lined by granulation tissue



Keloid

- Is a protuberant scar covered by stretched epidermis
- Due to overdone repair in susceptible persons e.g. Negros
- Recur after surgical excision → shrinkage by irradiation.



Cosmetic deformities and functional troubles

Contracture'
following a major
burn



Epidermal cyst (Implantation cyst)

- Cyst filled with keratin
- Due to proliferation of epithelial cells that were trapped within the depth of the wound during epithelial regeneration.



